DC Motor Speed Control Using PWM with Obstacle Detection

1. Abstract

This project demonstrates the control of a DC motor using **Pulse Width Modulation** along with obstacle detection using an **HC-SR04 ultrasonic sensor**. The system adjusts motor operation based on distance measured in real time. When an object is detected within a threshold distance (20 cm), the motor stops automatically. This ensures safety and efficient control in embedded applications.

2. Introduction

DC motors are widely used in automation, robotics, and industrial systems. Controlling their speed efficiently is important for improving performance and energy utilization.

PWM is a technique used to vary motor speed by adjusting the duty cycle of the signal instead of changing voltage directly. Additionally, obstacle detection using ultrasonic sensors enhances system intelligence by preventing collisions.

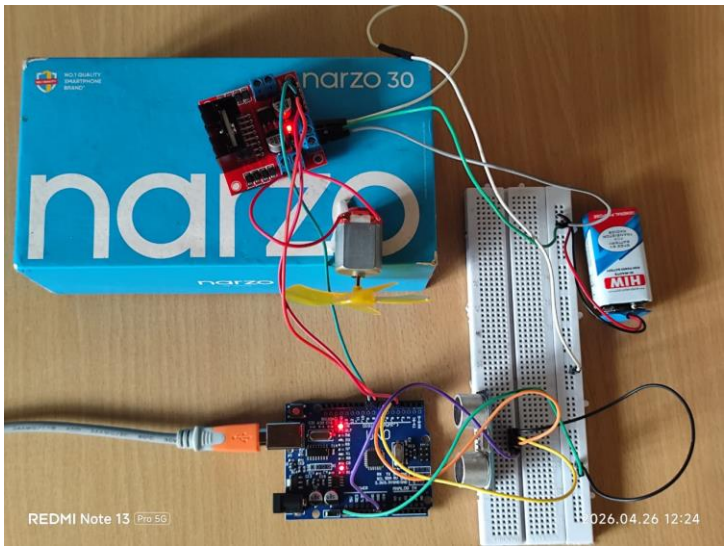
3. Objectives

- To control DC motor speed using PWM
- To measure distance using an ultrasonic sensor
- To stop the motor when an obstacle is detected
- To implement real-time embedded control

4. Components Required

- Arduino UNO / ESP32
- L293D motor driver IC
- DC Motor
- HC-SR04 ultrasonic sensor
- Power supply (Battery)
- Connecting wires

5. System Connection



6. Working Principle

◇ PWM Motor Control

PWM controls motor speed by varying the duty cycle:

- 0% → Motor OFF
- 50% → Medium speed
- 100% → Full speed

◇ Ultrasonic Sensor

The sensor works by transmitting ultrasonic waves and receiving the echo.

Distance formula:

*Distance=Time×Speed of Sound2Distance = \frac{Time \times Speed\ of\ Sound}{2}*Distance=2Time×Speed of Sound

◇ Control Logic

- Distance > 20 cm → Motor ON
- Distance ≤ 20 cm → Motor OFF

7. Arduino Code

```
#define ENA 5
#define IN1 8
#define IN2 9

#define TRIG 10
#define ECHO 11

long duration;
int distance;

int speedValue = 180;
int threshold = 20;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  Serial.begin(9600);
}

void loop() {

  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  duration = pulseIn(ECHO, HIGH);
  distance = duration * 0.034 / 2;

  if (distance > threshold) {
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);
```

```
    analogWrite(ENA, speedValue);

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm | MOTOR: ON");
  }
 else {
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
    analogWrite(ENA, 0);

    Serial.print("Distance: ");
    Serial.print(distance);
    Serial.println(" cm | MOTOR: OFF");
  }

  delay(200);
}
```

8. Experimental Results

Distance (cm)	Motor Status
35	ON
30	ON
25	ON
21	ON
15	OFF
12	OFF
10	OFF
9	OFF

9. Observations

- Motor runs smoothly when no obstacle is present
- Motor stops immediately when obstacle is near
- Threshold distance works effectively (~20 cm)
- Minor delay due to sensor processing

10. Advantages

- Simple and low-cost system
- Energy-efficient motor control
- Improves safety
- Easy implementation

11. Applications

- Obstacle avoiding robots
- Automatic vehicles
- Industrial safety systems
- Conveyor belt control

12. Limitations

- Ultrasonic sensor affected by noise
- Limited range
- Slight delay in response

13. Future Enhancements

- Implement PID speed control
- Use ESP32 for high-resolution PWM
- Add LCD/OLED display
- Add Bluetooth/WiFi control
- Implement gradual braking

14. Conclusion

The project successfully demonstrates DC motor speed control using PWM and obstacle detection using an ultrasonic sensor. The system responds accurately to distance changes and ensures safe motor operation, making it suitable for embedded and automation applications.